COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
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ACTIVITY NO. 11 **Containerization**

1. PROCEDURE OUTPUT

1. **Create a new repository for this activity.**

```
localserver@workstation:~$ mkdir -p ~/ansible-docker-project
localserver@workstation:~$ cd ~/ansible-docker-project/
localserver@workstation:~/ansible-docker-project$ git init
```

- Initializing a dedicated Git repository sets up local version control for tracking containerization files and configuration scripts.

2. **Install Docker and enable the docker socket.**

```
localserver@workstation:~/ansible-docker-project$ sudo systemctl enable --now docker.socket
localserver@workstation:~/ansible-docker-project$ sudo systemctl start docker
localserver@workstation:~/ansible-docker-project$ systemctl docker status
Unknown command verb 'docker', did you mean 'cancel'?
localserver@workstation:~/ansible-docker-project$ systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-10-13 17:08:29 PST; 2min 25s ago
 Invocation: 15911c20588644b6a42a84495e72cdb3
TriggeredBy: ● docker.socket
```

- Enabling the Docker socket and starting the Docker engine activates the container runtime service on the host system.

3. **Add to Docker group to your current user.**

```
localserver@workstation:~/ansible-docker-project$ sudo usermod -aG docker $USER
localserver@workstation:~/ansible-docker-project$ newgrp docker
localserver@workstation:~/ansible-docker-project$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

- Adding the current user to the Docker group grants permissions to manage Docker containers without administrative sudo privileges.

4. **Create a Dockerfile to install web and DB server.**

```
localserver@workstation:~/ansible-docker-project$ cat Dockerfile
cat << 'EOF' > Dockerfile
FROM ubuntu:latest

ENV DEBIAN_FRONTEND=noninteractive

RUN apt-get update && apt-get install -y \
        apache2 \
        mariadb-server \
        && apt-get clean \
        && rm -rf /var/lib/apt/lists/*

EXPOSE 80 3306

CMD service mariadb start && apachectl -D FOREGROUND
EOF
```

- Defining the Dockerfile specifies base image requirements and packages to combine the Apache web server and MariaDB database into a single container image.

5. **Install and build the Dockerfile using Ansible.**

```
localserver@workstation:~/ansible-docker-project$ pip3 install docker --break-system-package
s 2>/dev/null || pip3 install docker
Defaulting to user installation because normal site-packages is not writeable
```

```
localserver@workstation:~/ansible-docker-project$ ansible-galaxy collection install communit
y.docker
Starting galaxy collection install process
Nothing to do. All requested collections are already installed. If you want to reinstall the
m, consider using `--force`.
```

```
localserver@workstation:~/ansible-docker-project$ cat playbook.yml
cat << 'EOF' > playbook.yml
---
- name: Build Docker image for Web and DB server
  hosts: localhost
  connection: local
  tasks:
    - name: Build Docker image from Dockerfile
      community.docker.docker_image:
        build:
          path: "{{ playbook_dir }}"
        name: web-db-server
        tag: latest
        source: build

EOF
```

```
localserver@workstation:~/ansible-docker-project$ ansible-playbook playbook.yml

PLAY [Build Docker image for Web and DB server] ********************************************

TASK [Gathering Facts] *********************************************************************
ok: [localhost]

TASK [Build Docker image from Dockerfile] **************************************************
changed: [localhost]

PLAY RECAP *********************************************************************************
localhost                  : ok=2    changed=1    unreachable=0    failed=0    skipped=0
rescued=0    ignored=0
```

- Running the Ansible playbook automates building the local container image from the Dockerfile without manual intervention.

6. **Add, commit and push it to your repository.**

```
localserver@workstation:~/ansible-docker-project$ git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 636 bytes | 318.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:mo-na-min/ansible-docker-project.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
localserver@workstation:~/ansible-docker-project$
```

- Committing and pushing the files saves the project configuration history and uploads it to the remote GitHub repository.

GitHub Repository: https://github.com/mo-na-min/ansible-docker-project

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:

Answer the following:

1. What are the benefits of implementing containerizations?
   - Virtual machines need their own guest operating system and hardware so the setup stays fully separate. Containers do not work like that. They use the host machine's OS kernel, while still keeping each app in its own isolated space. Because of this, containers can come up very fast. They also take less RAM and use less disk space. Changes are easier too, since you can adjust the setup using config files such as Dockerfiles instead of doing a complete operating system upgrade.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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